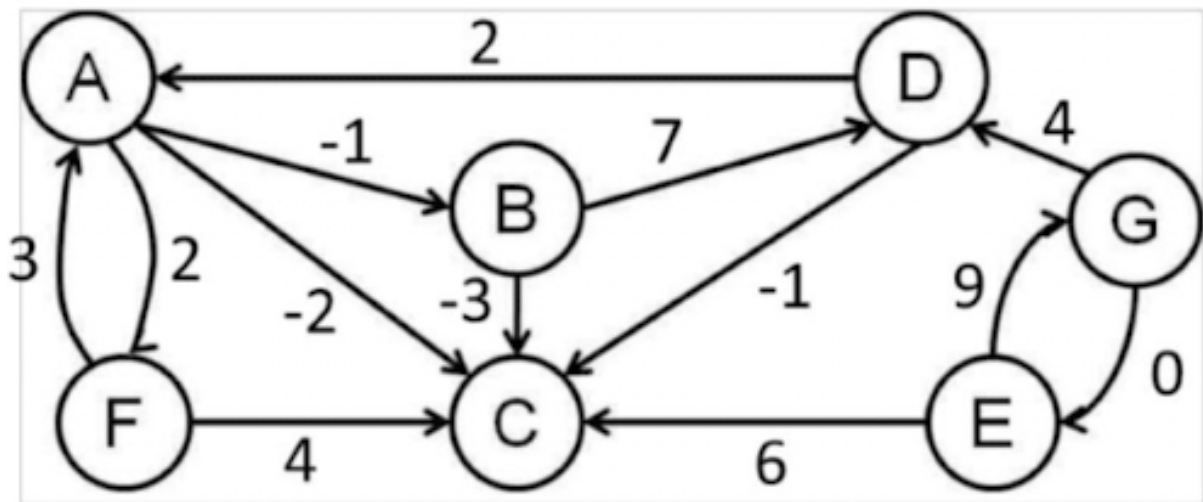


HW5

Algorithms Analysis 03

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Given a graph.



Fill the adjacency matrix below.

	A	B	C	D	E	F	G
A	0	-1	-2	∞	∞	2	∞
B	∞	0	-3	7	∞	∞	∞
C	∞	∞	0	∞	∞	∞	∞
D	2	∞	-1	0	∞	∞	∞
E	∞	∞	6	∞	0	∞	9
F	3	∞	4	∞	∞	0	∞
G	∞	∞	∞	4	0	∞	0

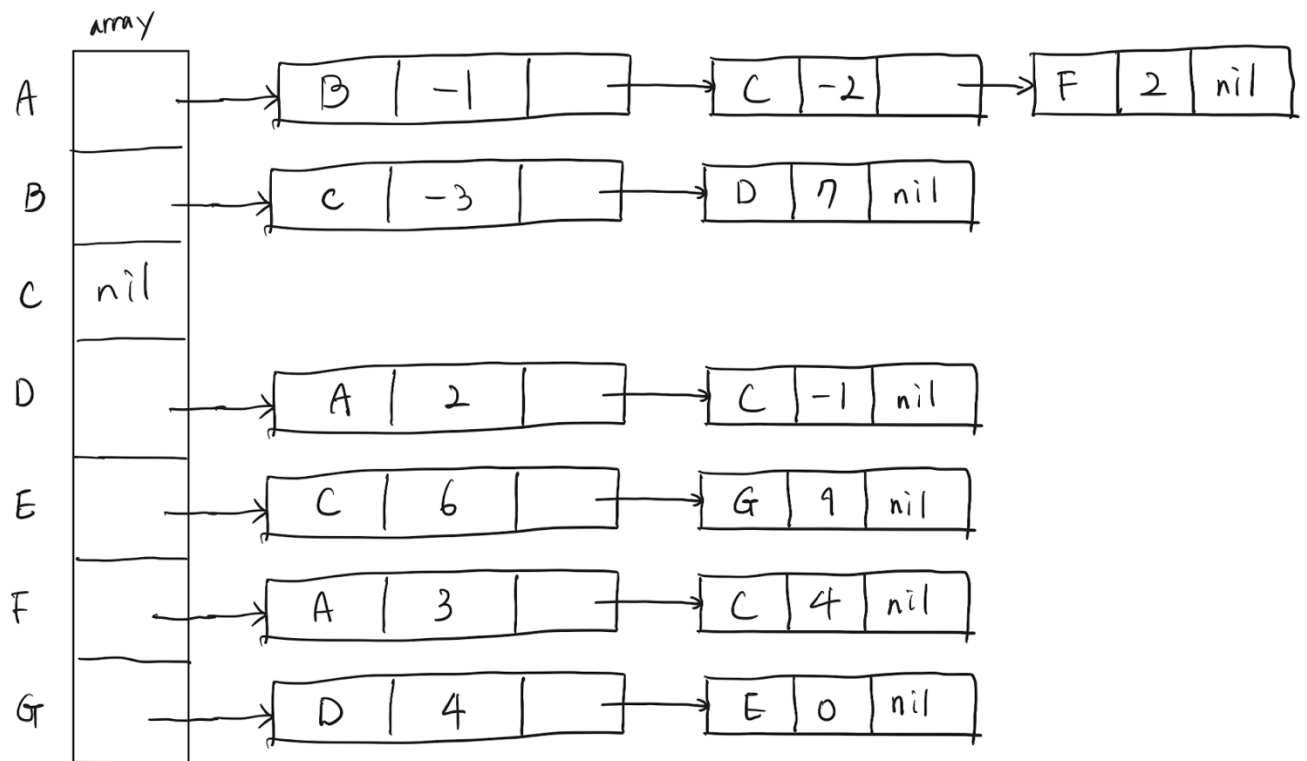
↳ weight = 0

↳ There is no self-loop edge.

G → (E, weight = 0)

Vertices like (A,A), ..., (G,G) are not included in the adjacency list, not because their weight is 0, but because there are no self-loop edges.

Write(Draw) the array of adjacency list in alphabetical order.



Vertex **C** has no outgoing edges.

All vertices other than C are connected to at least one neighbor.

Vertex **A** has outgoing edges to B, C, and F.

Vertex **B** has outgoing edges to C, and D.

Vertex **D** has outgoing edges to A, and C.

Vertex **E** has outgoing edges to C, and G.

Vertex **F** has outgoing edges to A, and C.

Vertex **G** has outgoing edges to D, and E.

And Each node in the adjacency list stores a weight value for that specific edge.

Transparency

- i. I verified through AI whether it is correct to add it to the adjacency list even when the weight is 0.
- ii. <https://chatgpt.com/share/69ec4003-579c-8322-9778-224988191fc4>